

Judge Q&A — Everything They Might Ask

SaResQ · answers, evidence and analogies · read this on the way in

Where every file is

WHAT	PATH
Deck script — 12 slides + 3 reserve, with what to say	~/saresq/docs/SaResQ_Deck_Draft.pdf
This Q&A sheet	~/saresq/docs/SaResQ_QA_Prep.pdf
Full technical report (19 pp, with glossary)	~/saresq/docs/SaResQ_Model_Architecture_Report.pdf
3D interactive model	claude.ai artifact — docs/assembly_3d.html
Build guide	claude.ai artifact — docs/assembly_guide.html
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If you remember one paragraph, remember this

"Drone thermal search already works — what fails is the human watching the feed. We put the watching onboard: a 3.2 MB model on a ₹4,000 Raspberry Pi that finds 85% of people in aerial thermal imagery, in real time, using 29% of its frame budget. It bolts to any 450 mm quadcopter and never touches the flight controller."

A · The problem, and why it matters

Q Why does this need to exist? Drones with thermal cameras already do this.

They do — and every one of them streams video to a human who has to watch it. That person gets tired. A survivor is a 2.6-pixel warm smudge on a 32×24 thermal image at 4 frames a second. They can't rewind, can't combine three weak sightings from three passes, and can't put a GPS coordinate on something they half-saw. We automate the *looking*, not the flying.

Q Who actually uses this? Who pays for it?

State disaster response forces and municipal fire services — organisations that already own drones and already fly them at incidents. The pitch is not "buy a drone"; it's "make the drone you own able to search on its own for ₹14,000."

That is the point of the zero-electrical-integration design.

Q What's the real-world impact?

Coverage that is **quantified rather than guessed**. Because the system models its own search coverage, it can report how thoroughly an area was actually searched — which is what lets a coordinator decide to move the team on instead of hoping. A human operator cannot give you that number.

B · "Isn't this just...?" — differentiation

Q Isn't this just YOLO running on thermal images?

If it were, it wouldn't run. A whole-frame detector needs about 320 ms per frame on a Pi 4 — the frame budget is 250 ms, so it's over budget before it does anything else. The contribution is the **cascade**: a statistical gate looks at every frame for free and nominates 2–6 small patches; the network only ever sees those.

Measured: 177 ms worst case, 29% headroom. That's the whole system in one number.

ANALOGY Airport security. You don't pat down everybody — everyone walks through a cheap metal detector, and only the few who beep get the expensive search. Our thermal gate is the metal detector.

Q Why not just buy a DJI thermal drone?

A DJI M30T is roughly ₹10–15 lakh and still **streams to an operator**. It's a better camera attached to the same bottleneck. We're a ₹14,000 module that adds autonomous perception to drones a department already owns. Different product, different price bracket, different problem.

Q What is genuinely new here?

Three things, and none of them is "we trained a model":

- **The cascade makes it fit.** Thermal-first nomination, then confirmation — $\sim 1/16$ th of a whole-frame detector's work.
- **Evidence accumulates across passes.** Three weak sightings of the same ground coordinate compound into one confident detection via a log-odds ledger. A live video feed can't do that at all.
- **The gate is two-sided.** Cold anomalies count — a body in water, a void under rubble. In afternoon sun, debris is *hotter* than a human, and every "find hot spots" system fails exactly then.

C · The AI and the tech stack

Q What's your tech stack?

LAYER	CHOICE	WHY
Detector	YOLOv8-nano, P3–P5 head	3.0 M params, 8.19 GFLOPs
Scene classifier	MobileNetV2, transfer-learned	ImageNet backbone + new head
Fusion	Logistic regression, 18 features	Readable weights, calibrated output
Training	PyTorch / Ultralytics, Apple M1 GPU (MPS)	Local — no cloud dependency
Deployment	LiteRT (TFLite) + NumPy	3 MB runtime vs 600 MB TensorFlow
Compute	Raspberry Pi 4, 1 GB, Debian 13	₹4,000
Sensors	MLX90640 · Pi Camera v2 · MPU-6050 · NEO-6M	all I ² C/UART, no HAT

No torch, no TensorFlow and no Ultralytics on the Pi — by design. That is why it fits in 1 GB.

Q Why YOLOv8 and not YOLOv11, or a transformer like DETR?

Because it has to run on a Pi. Transformer detectors are excellent and far too heavy — attention cost grows with the square of the number of patches, and we have a 1 GB budget. Newer YOLO versions are marginal gains at this scale; the ecosystem maturity of v8 (a stable, tested INT8 export path) mattered more than a point of mAP. That turned out to be the right instinct — see the quantisation question below.

Q You used pretrained weights. Isn't that cheating?

It's standard, and with 2,000 training images it's the only sensible choice. The detector starts from COCO, the classifier from ImageNet. We then trained the detector fully on thermal data — the whole network, not just a head.

ANALOGY Someone who can already read learns a new language far faster than someone learning to read from scratch. We didn't teach it what an edge is; we taught it what a human looks like in thermal, from above.

Q How much data? Isn't that too little?

Three datasets: **HIT-UAV** (2,029 thermal aerial), **RGBTDronePerson** (4,900 thermal, night urban) and **AIDER** (6,433 disaster scenes). The detector trains on **6,929 combined thermal images**. That's small for training from scratch and normal for fine-tuning a pretrained backbone.

Q How do you know it hasn't just memorised the dataset?

Because we test on a second, independently collected one. The model scores 0.850 on HIT-UAV and 0.544 on RGBTDronePerson — different sensor, different country, different time of day, different camera angle. A memoriser scores near zero on the second.

Before we added the second dataset, it scored 0.275 there. Training on both cost nothing on the first.

Q What about false positives?

Three defences. **(1)** The gate is deliberately permissive — in rescue a miss is far worse than a false alarm, so we push rejection downstream. **(2)** The detector confirms every nomination. **(3)** The fusion ledger requires *repeated* evidence at the same ground coordinate across passes, so a one-off hot roof tile never accumulates.

D • Defending the numbers

Q 85% — is that good?

For a **3.2 MB model on a Raspberry Pi**, yes. It is not state-of-the-art, and we don't claim it is. A model fifty times larger on a GPU would beat it. The constraint is the contribution: this has to run on hardware a relief agency can buy fifty of.

SAY IT PLAINLY "Person AP50 of 0.850. That means it finds and correctly boxes about 85 of every 100 people in the imagery it was built for."

Q What exactly does AP50 mean?

A prediction counts as correct if its box overlaps the real person by at least 50%. AP then summarises the whole precision-recall curve into one number. **AP50 is the standard detection metric** — it is not the same as "accuracy", which only makes sense for classification.

Q What's your AP50-95? (the sharp question — expect it from a strong judge)

0.371, and here's the honest reading: it finds people reliably and boxes them loosely. At IoU 0.75 it drops to 0.234.

For this mission that's the right trade. We need *"is there a person, and at what ground coordinate"* — not a traced silhouette. At 14.7 mm per pixel, a box 5 px loose is **7 cm on the ground**. No rescue outcome depends on 7 cm. In a medical segmentation system I'd call 0.371 a defect; here it's capacity spent on size and speed instead.

Volunteering this before they find it is worth more than the number costs you.

Q Is that measured on the model you actually deploy, or on your laptop?

Both — and they match. The 3.2 MB file that runs on the Pi scores 0.860; the PyTorch reference scores 0.850. We built a test that compares them directly, because that comparison caught a serious bug.

Q What was the bug?

Standard 8-bit quantisation was silently running our model at 32% of its real accuracy. 0.2315 instead of 0.8229. It loaded fine, ran fast, and drew sensible-looking boxes. Nothing threw an error.

Cause: full INT8 quantises the *activations* too, including the box head's internal 1/640 normalisation constant — which came out as an effective 1/636. On people 9–45 px tall a 1–2 px shift destroys an IoU-0.5 match. We switched to **w8a32** — 8-bit weights, float activations. Same 3.3 MB file, **99.2%** of reference.

ANALOGY Like saving every photo as a heavily compressed JPEG. On screen it looks fine. Measure it and a third of the detail is gone. We only knew because we measured.

Q Why should we believe your numbers at all?

Every figure is measured on this hardware and reproducible from one command (`training/run_all.sh`). **167 automated tests pass.** And the numbers include the ones that told us we were wrong — we report the architecture we rejected and the bug we shipped against ourselves.

E · Architecture choices

Q Why P3–P5 and not P2? Small objects need high resolution.

We built both and measured. P2 lost every object-size bin at twice the compute.

BIN	P3	P2
8–16 px	0.657	0.612
16–32 px	0.772	0.753
32+ px	0.814	0.791
GFLOPs	8.86	17.43

Three reasons: resolution was never the bottleneck at 8–32 px; four times the anchor points makes optimisation harder for no extra signal; and the deployed path is a 160 px *crop*, where a prone adult is 115×31 px — not a small-object problem at all.

ANALOGY Reaching for a stronger magnifying glass when the problem was never magnification. It cost twice the effort and saw less.

Q What do P2, P3, P4, P5 mean?

Feature maps at different scales. **P_n = the image shrunk by 2ⁿ**. P3 is 1/8 size (80×80 from 640 px), P5 is 1/32. Fine levels catch small objects, coarse levels catch big ones. Adding P2 (1/4 size) quadruples the area of that layer — which is where the compute went.

Q Why a thermal camera with only 32×24 pixels? That's nothing.

Because it never has to identify anybody. At 20 m a prone adult is 2.1 thermal pixels — there's no texture to learn, only *contrast*. So the thermal stage is a statistical anomaly test, not a classifier. Identification is the visual camera's job, on the crop the thermal nominated. A radiometric core would be 20–40× the cost and would not change the architecture.

ANALOGY A smoke detector doesn't identify what's burning. It just says "something here, go look." That's our thermal stage.

Q Why logistic regression for fusion? Why not a neural network?

Two reasons. **Calibration:** the ledger sums log-odds across passes, so the output must be an honest probability — a bias compounds instead of averaging out. Our expected calibration error is 0.0058. **Explainability:** a rescue coordinator can be told *why* a candidate was flagged, because the weights are readable numbers.

F · Hardware, deployment, real time

Q Why a Raspberry Pi and not a Jetson Nano?

Cost and power — and power is flight time. A Jetson is 5–10× the price and draws far more. **Anyone can run a large model on a large computer; that isn't deployable at a disaster-response budget.** Our whole process uses 95 MB of RAM and 3.2 MB of model. A state agency can buy fifty.

Q Prove it runs in real time.

Measured on the board, not extrapolated:

OPERATION	MEDIAN
Detector, 160 px crop	28.2 ms
Detector, 224 px crop	56.7 ms
Hazard classifier, full frame	33.0 ms
Worst case, 6 crops + hazard	177 ms
Budget at 4 Hz thermal	250 ms

95.2 MB RAM of 905 available · 42.8 °C · no throttling · on the payload's own power bank.

Q How does it mount to a drone?

Four M3 bolts into a standard vibration-damping plate, or two straps through built-in slots. **340 g total**, suits a 450 mm-class quadcopter (F450/S500) or bigger. **Critically, it has zero electrical connection to the flight controller** — own battery, own GPS, own IMU. It can't brown out the flight stack or cause a flyaway, so fitting it is a mechanical job, not an avionics integration.

Q Why does the damping plate matter?

It isn't comfort — it's detection accuracy. The IMU's attitude solution is what converts a detected pixel into a GPS coordinate. An undamped IMU on a quadcopter gives attitude noise, which becomes position error on every single detection.

Q What altitude does it work at?

20–30 m, optimum about 25 m — and the limit is the sensor, not the AI. Below ~30 m a person fills at least one whole thermal pixel, so you get full contrast. Above that they only partly fill a pixel and their heat averages with the cold ground — apparent contrast falls as $1/\text{height}^2$. At 25 m you get a 26 m swath. The detector barely degrades across that range; the thermal sensor does.

Q What about direct sunlight? Rain? Night?

Night is our best case — maximum thermal contrast, and the visual camera is useless anyway. **Direct afternoon sun is the worst case:** sun-baked concrete can be hotter than a human, which is exactly why the gate is two-sided rather than "find hot spots." We recommend dawn, dusk or night. **That is a physics constraint, and no software fixes it.** Rain degrades any thermal system; we make no claim there.

Q Battery life? What if it crashes mid-flight?

The payload runs from its own 10,000 mAh bank, independent of the airframe's flight battery — so it outlasts the drone's endurance and **cannot drain it**. If the software crashes, the drone is unaffected, because the payload has no authority over it. That independence is deliberate.

G · Cost, scale, regulation

Q Break down the cost.

Roughly ₹10,000–14,000: Raspberry Pi 4 ~₹4,000 · MLX90640 thermal ~₹4,000 · Pi Camera ~₹1,200 · power bank ~₹1,000 · IMU + GPS ~₹800 · printed parts ~₹300 of filament · cabling and fasteners. **No licence fees, no cloud costs, no subscription** — it runs entirely offline, which also means it works where the network is down, which after a disaster is everywhere.

Q Can this scale to manufacturing?

Every part is commodity and in stock. The only custom item is the printed tray, which is parametric CAD — a design change is a one-line edit and a 12-second rebuild, and at volume it would be injection-moulded. Nothing here is bespoke or single-sourced.

Q DGCA regulations?

It's a passive payload with no flight authority, so it doesn't change the aircraft's certification basis — the operator flies under whatever rules already apply to their drone. **Be honest: we have not sought a formal opinion, and we'd do that before deployment.**

H • The hard questions — answer these honestly

Rule for this whole section

Say the weakness first, in plain words, then what you'd do about it. A judge who finds a flaw you hid stops trusting everything else you said. A judge you hand a flaw to starts trusting you more.

Q Have you actually flown this?

No. It's benchmarked on the target board, not in the air. Vibration, attitude dynamics and real thermal scenes over rubble are all unvalidated. That's the single biggest gap and we know it.

Q What doesn't work yet?

- **The fusion stage is weak** — ROC AUC 0.670. Five of its eighteen inputs need the sensors mounted and flying before they can be measured at all.
- **No daytime visible detector.** The system is thermal-only today.
- **Our search-coverage figure is withheld** — the only measurement available is out-of-domain, and publishing it would overstate coverage roughly 5×.

Q Why no visible-light detector? You have a camera.

We tried, and the dataset defeated us — for an interesting reason. Our RGB training set turned out to be **night-time imagery**: 32% of frames have a mean brightness under 40/255. The labelled people simply aren't visible in the visible channel.

The proof: our thermal model, untrained on that data, recalls **49.9%** of those labels on the thermal frames and **0.0%** on the visible ones. Zero matches from 692 detections isn't "difficult", it's "not there." No training configuration could have worked. It needs a daytime aerial dataset, which is scoped future work.

WORTH SAYING "We spent three training runs treating it as a model problem before we checked the data. The lesson — verify the signal exists before you touch hyperparameters — is the most useful thing we learned."

Q Can it see someone buried under rubble?

No — and no thermal camera can. Thermal is line-of-sight; it sees surface temperature. We detect people who are *exposed or partly exposed*: on the surface, in a void opening, at the edge of debris. Anyone claiming a thermal drone sees through concrete is wrong. For buried victims you need acoustic or radar sensing, which is a different instrument.

Q What's your biggest risk?

That gate recall in the real domain is worse than we hope. Our only measurement of it is on urban night scenes, which is out-of-domain for a contrast gate tuned for rubble. The fix is a measurement, not a redesign: point the sensor at a real debris field and count.

Q How much of this did you actually build?

Be specific and honest. Datasets and pretrained backbones are public. What's ours: the cascade architecture, the thermal gate, the fusion ledger, the deployment runtime, the quantisation study, the parametric CAD, and the validation harness — 167 tests. Say what was AI-assisted if it was; judges respect that more than a vague claim.

I • Numbers to have on the tip of your tongue

THING	NUMBER	CONTEXT
Person AP50, HIT-UAV	0.850	the headline
Person AP50, second dataset	0.544	proves generalisation
AP50-95	0.371	loose boxes, fine for rescue
Hazard classifier	95.2%	5 classes
Model size	3.2 MB	w8a32

Inference, 160 px crop	28.2 ms	on the Pi
Worst-case frame	177 ms	of a 250 ms budget
RAM used	95 MB	of 905 available
Payload mass	340 g	450 mm-class quad
Cost	₹14,000	bill of materials
Best altitude	25 m	26 m swath
Tests passing	167	

J • Four analogies that do a lot of work

FOR	SAY
The cascade	Airport security — everyone through the cheap metal detector, only the few who beep get searched.
The thermal gate	A smoke detector. It doesn't identify what's burning; it says "something here, go look."
The quantisation bug	Saving every photo as a heavy JPEG. Looks fine on screen; measure it and a third of the detail is gone.
Transfer learning	Someone who can already read learning a new language, versus learning to read from scratch.

The close, if you get one line

"Every number we showed you was measured on this hardware — including the ones that told us we were wrong."

All figures traceable to docs/SaResQ_Model_Architecture_Report.pdf · repository commit 9dc0569 · 167 tests passing.